Exercise 9-1: Analyze time-series data

Did you use an LLM?

No, I did not use LLM for this assignment, I followed instructions to perform the task, I check solutions to verify my answers

In [1]:
```
import pandas as pd
import seaborn as sns
```

In [2]:
```
stockData = pd.read_pickle('stocks.pkl')
stockData.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 253 entries, 0 to 252
Data columns (total 5 columns):
 #   Column  Non-Null Count  Dtype
---  ------  --------------  -----
 0   Date    253 non-null    datetime64[ns]
 1   Open    253 non-null    float64
 2   High    253 non-null    float64
 3   Low     253 non-null    float64
 4   Close   253 non-null    float64
dtypes: datetime64[ns](1), float64(4)
memory usage: 10.0 KB
```

In [3]:
```
# display the first five rows
stockData.head(5)
```

Out[3]:

	Date	Open	High	Low	Close
0	2020-01-02	74.059998	75.150002	73.797501	75.087502
1	2020-01-03	74.287498	75.144997	74.125000	74.357498
2	2020-01-06	73.447502	74.989998	73.187500	74.949997
3	2020-01-07	74.959999	75.224998	74.370003	74.597504
4	2020-01-08	74.290001	76.110001	74.290001	75.797501

Generate date ranges

In [4]:
```
# generate a daterange for every other day in the year 2020
pd.date_range('01/01/2020', '12/31/2020', freq = '2D')
```

Out[4]:
```
DatetimeIndex(['2020-01-01', '2020-01-03', '2020-01-05', '2020-01-07',
               '2020-01-09', '2020-01-11', '2020-01-13', '2020-01-15',
               '2020-01-17', '2020-01-19',
               ...
               '2020-12-12', '2020-12-14', '2020-12-16', '2020-12-18',
               '2020-12-20', '2020-12-22', '2020-12-24', '2020-12-26',
               '2020-12-28', '2020-12-30'],
              dtype='datetime64[ns]', length=183, freq='2D')
```

In [5]:
```
# generate a daterange for every 3 hours in 2020
pd.date_range('01/01/2020', '12/31/2020', freq = '3H')
```

Out[5]:
```
DatetimeIndex(['2020-01-01 00:00:00', '2020-01-01 03:00:00',
               '2020-01-01 06:00:00', '2020-01-01 09:00:00',
               '2020-01-01 12:00:00', '2020-01-01 15:00:00',
               '2020-01-01 18:00:00', '2020-01-01 21:00:00',
               '2020-01-02 00:00:00', '2020-01-02 03:00:00',
               ...
               '2020-12-29 21:00:00', '2020-12-30 00:00:00',
               '2020-12-30 03:00:00', '2020-12-30 06:00:00',
               '2020-12-30 09:00:00', '2020-12-30 12:00:00',
               '2020-12-30 15:00:00', '2020-12-30 18:00:00',
               '2020-12-30 21:00:00', '2020-12-31 00:00:00'],
              dtype='datetime64[ns]', length=2921, freq='3H')
```

In [6]:
```
# generate a date range for every other Friday in 2020
pd.date_range('01/01/2020', '12/31/2020', freq = '2W-FRI')
```

Out[6]:
```
DatetimeIndex(['2020-01-03', '2020-01-17', '2020-01-31', '2020-02-14',
               '2020-02-28', '2020-03-13', '2020-03-27', '2020-04-10',
               '2020-04-24', '2020-05-08', '2020-05-22', '2020-06-05',
               '2020-06-19', '2020-07-03', '2020-07-17', '2020-07-31',
               '2020-08-14', '2020-08-28', '2020-09-11', '2020-09-25',
               '2020-10-09', '2020-10-23', '2020-11-06', '2020-11-20',
               '2020-12-04', '2020-12-18'],
              dtype='datetime64[ns]', freq='2W-FRI')
```

Reindex the data

In [7]:
```
# set an index on the date column for the stock data
stockData.set_index('Date', inplace = True)
stockData.head()
```

Out[7]:

Date	Open	High	Low	Close
2020-01-02	74.059998	75.150002	73.797501	75.087502
2020-01-03	74.287498	75.144997	74.125000	74.357498
2020-01-06	73.447502	74.989998	73.187500	74.949997
2020-01-07	74.959999	75.224998	74.370003	74.597504
2020-01-08	74.290001	76.110001	74.290001	75.797501

In [8]:
```
# reindex the data so the data contains only Fridays
# and assign the result to a variable called stockDataFridays
```

```
stockDataFridays = stockData.reindex(pd.date_range('01/01/2020', '12/31/2020', f
stockDataFridays.head()
```

Out[8]:

	Open	High	Low	Close
2020-01-03	74.287498	75.144997	74.125000	74.357498
2020-01-10	77.650002	78.167503	77.062500	77.582497
2020-01-17	79.067497	79.684998	78.750000	79.682503
2020-01-24	80.062500	80.832497	79.379997	79.577499
2020-01-31	80.232498	80.669998	77.072502	77.377502

In [9]:
```
# use Pandas to plot the Close column of the reindexed data
stockDataFridays.plot(y = 'Close')
```

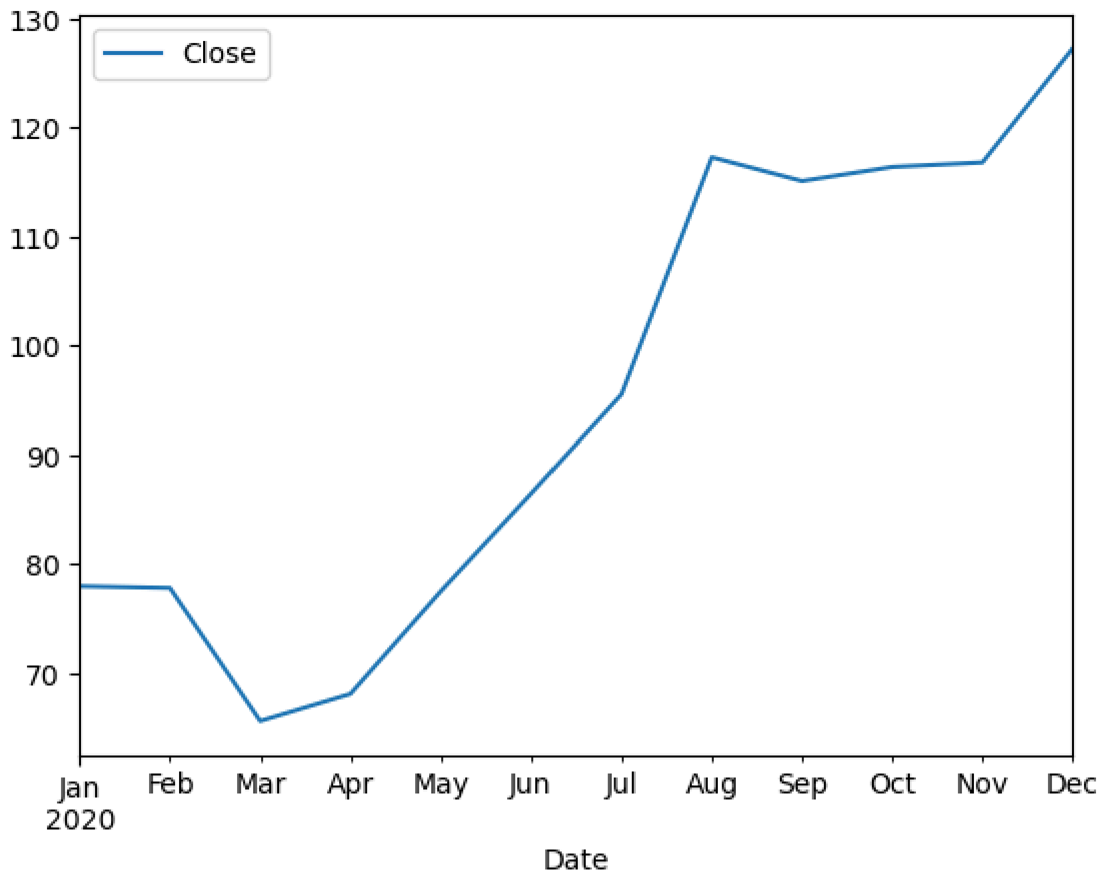
Out[9]: <Axes: >

Resample the data

In [10]:
```
# downsample the data to a monthly frequency
stockDataDown = stockData.resample(rule='M').mean()
```

In [11]:
```
# use Pandas to plot the Close column of the resampled data
stockDataDown.plot(y='Close')
```

Out[11]: <Axes: xlabel='Date'>



Compute a rolling window

In [12]:

```
# compute a 2 week rolling average for the Close column
# set the min_periods to 1 and assign the data to a variable called stocksRollin
stocksRolling = stockData[['Close']].rolling(window=14, min_periods=1).mean()
stocksRolling.head()
```

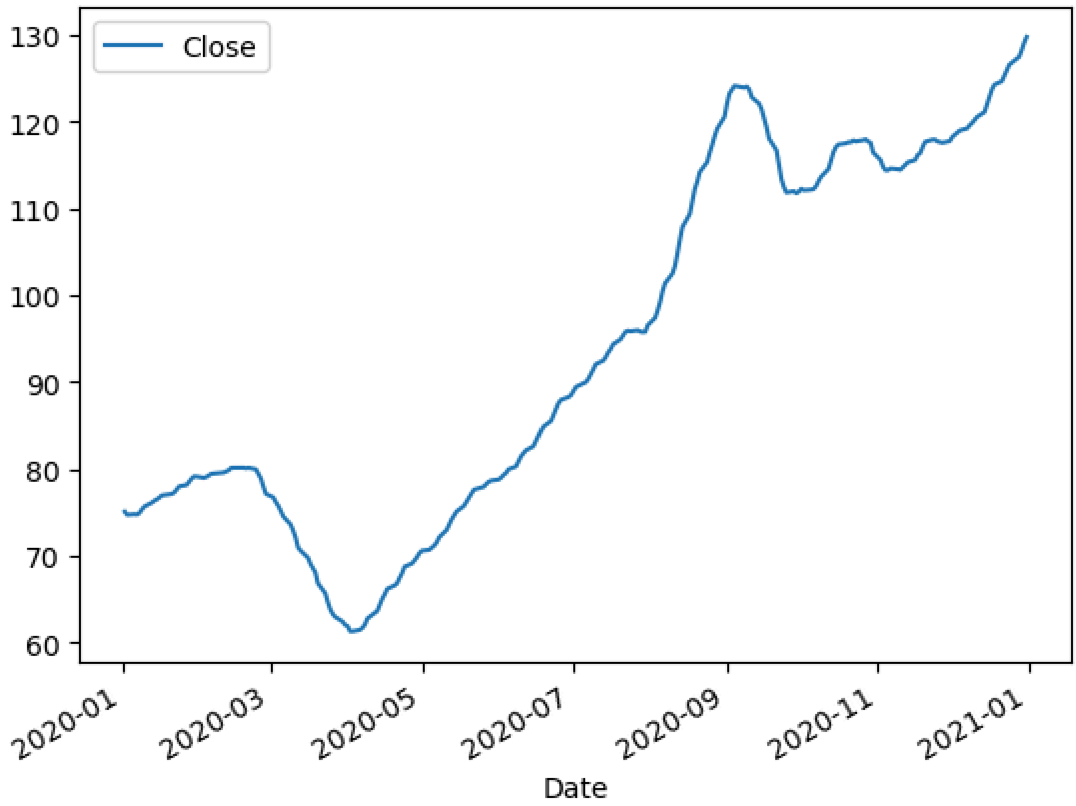
Out[12]:

	Close
Date	
2020-01-02	75.087502
2020-01-03	74.722500
2020-01-06	74.798332
2020-01-07	74.748125
2020-01-08	74.958000

In [13]:

```
# use Pandas to plot the Close column of the rolling data
stocksRolling.plot()
```

Out[13]: <Axes: xlabel='Date'>


Close
130
120
110
100
90
80
70
60
2020-01
2020-03
2020-05
2020-07
2020-09
2020-11
2021-01
Date

In [ ]: